

Certified but Compromised: Breaking and Fixing Lightweight Proofs of Inference

Final Report — Trustworthy Machine Learning, Spring 2026

Ofek Lavi
Tel Aviv University
Tel Aviv, Israel
ofeklavi@mail.tau.ac.il

Erel Barzilay
Tel Aviv University
Tel Aviv, Israel
erelbarzilay@mail.tau.ac.il

Edo Kadosh
Tel Aviv University
Tel Aviv, Israel
edo@mail.tau.ac.il

Abstract

A company that outsources inference is exposed to changes in the computation, made either to cut costs or to return a wrong answer on purpose. Proofs of inference let it check each answer. Those built on zkSNARKs give strong guarantees but take minutes per query on large models. This has motivated lighter protocols that check only a sample of the computation, such as that of Anchuri et al., where the client re-checks one random path through a committed execution trace. We show that their method misses targeted changes: a provider that tampers with only a single neuron almost always passes the path check, on image classifiers and on a real large language model. We prove that no sampling rule closes this gap on every network. We therefore create a protocol that verifies every layer using Freivalds’ algorithm against a Reed–Solomon/Merkle commitment to the weights. It needs no SNARK, trusted hardware or trusted setup, and a wrong output passes with negligible probability. Its prover is orders of magnitude faster than published zkSNARK provers and faster than every other fully sound system we compare with, and its verifier is faster than all of them but two, whose provers take minutes per query; its main cost is a proof that grows with the input.

Keywords

verifiable inference, proof of inference, integrity attacks, Freivalds’ algorithm, zkML

1 Introduction

Consider a company that pays a cloud provider to serve a licensed model to its users. Nothing in a response shows that the provider ran that model with the agreed weights. The provider may run a cheaper, different model, or tamper with the computation to return wrong answers to chosen queries, hurting processes that act on the model’s answers, such as payment approvals or fraud flagging. The company cannot re-run the model to find out, since it lacks the weights and the hardware; that is why it outsourced inference. A *proof of inference* closes this gap. With each answer the provider sends a proof, and the client runs a check, much cheaper than inference, that fails with high probability unless the answer was computed with the agreed weights on this input.

zkSNARKs [9, 16, 26, 30, 32] prove that the answer is correct and can keep the weights secret. Their proofs are usually small and fast to check, but producing one takes significantly longer than inference; zkLLM needs about ten minutes on an A100 for one Llama-2-7B prompt [32]. Due to the nature of the problem, we attempt to improve mainly prover time, as this is the main bottleneck for

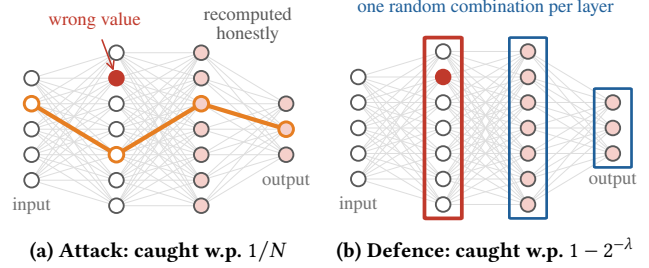


Figure 1: One tampered neuron slips past a random path, but not past a check of every layer

protocols that provide full soundness. Anchuri et al. [2] proposed a much lighter approach. The provider commits to the full execution trace with a Merkle tree, and the client checks the computation along *one random path* from an output neuron back to the input, against a commitment to the weights. Proving drops from minutes to milliseconds. A different model produces a trace that differs from the correct one in many places, and a random path is likely to hit one of them. Anchuri et al. accordingly prove *other-model* soundness, against a provider that honestly runs a different model. They note that one path cannot catch a single changed node with probability above $1/N$ in a layer of width N , and they leave *full* soundness, against any wrong answer, open.

We present an attack on this protocol. A provider that keeps the committed model and changes only what it needs to can overwrite *a single neuron* and recompute everything after it honestly (Figure 1a). The trace is then inconsistent at a single node, which a random path visits with probability $1/N$, so almost every tampered answer is accepted. Applied only to inputs that carry a trigger, the edit is a backdoor in the serving code, which leaves the committed weights clean and answers clean queries bit for bit as before. Checking many queries does not close the gap, because the provider chooses which queries to tamper with, and tampering with k of them is caught with probability at most k/N . The fixes that Anchuri et al. suggest, sampling more paths or sampling where it matters, prove unhelpful: no rule does better than uniform sampling on every network, and reaching 2^{-40} forces the path protocol to open essentially the whole model and trace.

This led us to a different design (Figure 1b). The client tests a few random linear combinations of all the neurons of every layer (Freivalds’ algorithm [15]). When it does not hold the weights, it checks the combination against a Reed–Solomon-encoded, Merkle-committed copy of them in the style of Ligerio [1], so the

Table 1: Our costs against published systems: our prover always wins (in parentheses: ours lower ↓ or higher ↑, bold if lower)

System	Model (tokens)	prover↓		verifier↓		proof↓	
		theirs	ours	theirs	ours	theirs	ours
zkCNN ^a	LeNet-5	441 ms	5.7 ms (78×↓)	5.8 ms	2.8 ms (2.1×↓)	71.3 kB	63.3 kB (1.1×↓)
Bionetta	LeNet-5	3.8 s	5.7 ms (664×↓)	10.0 ms	2.8 ms (3.6×↓)	880 B	63.3 kB (72×↑)
zkCNN ^a	VGG-16	88.3 s	20.0 ms (4,422×↓)	59.3 ms	20.9 ms (2.8×↓)	341 kB	2.2 MB (6.4×↑)
ZKML	VGG-16	10.6 min	20.0 ms (31,910×↓)	9.6 ms	20.9 ms (2.2×↑)	12.1 kB	2.2 MB (180×↑)
zkGPT ^b	GPT-2 (64)	21.8 s	95.0 ms (229×↓)	350 ms	96.8 ms (3.6×↓)	101 kB	16.6 MB (164×↑)
DeepProve ^c	GPT-2 (64)	34.2 s	95.0 ms (360×↓)	1.4 s	96.8 ms (14×↓)	21.7 MB	16.6 MB (1.3×↓)
DeepProve ^c	GPT-2 (512)	2.9 min	177 ms (995×↓)	1.6 s	437 ms (3.8×↓)	25.5 MB	91.2 MB (3.6×↑)
ZKTorch	Llama-2-7B (1)	44.1 min	2.2 s (1,213×↓)	1.7 min	333 ms (301×↓)	22.9 MB	190 MB (8.3×↑)
Maverick ^d	Qwen3-4B (8)	354 ms	117 ms (3.0×↓)	260 ms	111 ms (2.3×↓)	36.1 MB	35.2 MB (1.03×↓)
zkLLM	OPT-125M (2,048)	73.9 s	396 ms (187×↓)	342 ms	318 ms (1.1×↓)	141 kB	348 MB (2,467×↑)
zkLLM	OPT-1.3B (2,048)	3.7 min	3.8 s (58×↓)	899 ms	2.6 s (2.9×↑)	147 kB	2.1 GB (13,979×↑)

^aVerifier on one core (ours: 8 threads). ^bNo prompt length stated (its matrix shapes imply 32 tokens); ours: 64. ^cBaseFold configuration, proving every position. ^dOurs in K_{pre} , $\lambda=40$, one thread, as Maverick; without the compact encoding; Maverick’s prover runs on 64 CPU threads. Setting in Section 4.6.

client stores only a 32-byte Merkle root per weight matrix. The model runs as an int8 network whose results are identical on the provider’s GPU and on the client, which makes the test exact. A tampered value anywhere in the network is then caught with probability $1 - 2^{-\lambda}$ instead of $1/N$, and the provider’s extra work grows with the model but not with the input. At 2^{-128} our proofs are also smaller than the paths’ at 2^{-40} : catching the single-neuron attack on a 64-token Llama-2-7B prompt with probability $1 - 2^{-40}$ takes paths that open about 13.7 GB, while our proof is 325 MB.

Contributions. (1) **An attack** (Sections 3.2 and 4.2) that exploits the $1/N$ limit stated in [2]. One tampered neuron is accepted with probability $1 - 1/N$ (99.6–99.8% on MNIST) and changes the next token of 40 of 40 prompts on the real OPT-6.7B. (2) **A bound on sampling-based fixes** (Section 3.3): no sampler beats uniform sampling on every network, and uniform sampling requires essentially the whole model and trace for 2^{-40} . (3) **A full-soundness protocol** for whole CNNs and transformers (Sections 3.4–3.6), with error $2^{-\lambda}$, that composes known checks and needs only a commitment per weight matrix (Section 2). Optimisations keep the bound and shrink the committed-weights proof 2.0–4.2× on the models of Section 4.5. (4) **A matched-model evaluation** (Section 4) on six image classifiers and 10 LLM architectures up to 13B parameters, against [2] and 15 published systems. In Table 1 our prover is 58–32,000× faster than the zkSNARK provers and faster than every other fully sound system; only ZKML and zkLLM have faster verifiers than ours, and their own provers take minutes per query.

2 Related Work

zkSNARK proofs of inference. A long line of work compiles neural networks into zero-knowledge proofs. For CNNs these include sum-check/GKR provers [18] such as zkCNN [26] and zkPyTorch [36], and pairing- or Plonk-based systems such as vCNN [24], ZEN [12], ZENO [13], ZKML [9], EZKL [41] and Bionetta [39]; Mystique [35] is an interactive, VOLE-based alternative. For LLMs they include zkLLM [32] (also interactive), zkGPT [30], DeepProve [16], ZKTorch [8] and Jolt Atlas [5]. They give computational soundness and mostly small proofs, but their

provers are far slower than inference: from half a second for LeNet-5 and tens of seconds for GPT-2 to tens of minutes for 7B models. For brevity we call all of them zkSNARKs, although Mystique and zkLLM are interactive.

Interactive and Freivalds-style checks. SafetyNets [17] uses interactive proofs for networks with quadratic activations. Slalom [34] checks every linear layer of a GPU computation from inside an SGX enclave with Freivalds’ algorithm, optionally with a precomputed secret challenge; TwinShield [37] and VeriAttn [10] extend this to attention inside TEEs. Several recent works do not require an enclave. Maverick [27] checks every linear layer of a language model with sparse Freivalds checks on a code-encoded copy of the weights that the client stores, drawing a fresh challenge per query. Gupta et al. [20] run Freivalds’ check in the exponent of a group, against verification data that the model owner generates with secret randomness, for a client that outsources inference to two non-colluding servers. CommitLLM [7] checks int8 language models with secret Freivalds vectors precomputed from the weights, audits sampled positions and layers, and leaves attention unverified. LAMP [23] lets a verifier that holds only a commitment check one matrix product, using Freivalds’ check and Reed–Solomon proximity testing inside a Groth16 SNARK.

Sampling, fingerprints and optimistic protocols. Anchuri et al. [2] spot-check random paths and argue security through trace separation; SLP [25] proves a sampled subset of trace chunks with a SNARK. TOPLOC [29] and SVIP [33] fingerprint hidden states, with empirical guarantees. Verde [3], opML [11] and TAO [38] are optimistic: an answer stands unless a party that re-runs the model disputes it, and a dispute is narrowed down to one operation that a referee re-executes. They thus rely on an honest party, such as a second provider or a challenger, that re-runs every query.

Where we fit. We keep the setting of [2], a client that holds only a commitment to the weights, but replace its path test, which we prove does not provide full soundness $2^{-\lambda}$ (Sections 3.2 and 3.3), with a check of every layer that does, for complete CNNs and transformers (Sections 3.4 and 3.5). We chain known protocols through every layer with an end-to-end soundness proof: Freivalds’ check, and the column check of the Ligero and Brakedown commitments [1, 19]. The protocols above that avoid a SNARK each need

more than a cryptographic commitment: Slalom an enclave, Maverick an encoded copy of the weights, CommitLLM a secret key derived from them, and Gupta et al. public data that grows with every matrix and is generated with secret randomness. zkSNARKs that verify against a Merkle root, such as DeepProve’s BaseFold configuration [40], need no trusted setup but prove far more slowly. To our knowledge, ours is the first protocol that is fully sound for complete CNNs and transformers using only a hash function and a linear code, with no SNARK, TEE or trusted setup.

3 Technical Approach

3.1 Setting and threat model

A model M has a public architecture; the client may not hold its weights. The provider acts as the prover and the client as the verifier. Before any query the client obtains a commitment C_M to the weights, for example from the model owner. As in [2], we assume that C_M is computed honestly; unlike a SNARK’s trusted setup it involves no secret randomness, and anyone who holds the weights can recompute it. For each query x the provider returns a claimed output y and a proof, and may deviate from the protocol arbitrarily. We require *completeness* (honest answers are always accepted) and *full soundness*: any $y \neq M(x)$ is rejected with probability at least $1 - \epsilon$ (Anchuri et al. prove only other-model soundness). We consider three trust settings. In setting C the client holds only a short commitment C_M , as in [2] and the zkSNARKs (ours: 32 bytes per weight matrix); in K it holds the weights; and in K_{pre} it also precomputes secret Freivalds vectors and their folded rows (Eq. (1)) once per model, following Slalom’s preprocessing mode and CommitLLM [7]. The architecture includes the integer constants of the operations without weights (requantisation multipliers, look-up tables), which the client obtains with C_M .

3.2 The path test and the single-neuron attack

In the protocol of [2] (RandPathTest) the prover commits to every activation a_j of the forward pass with a Merkle tree. The verifier chooses a random output neuron and walks back to the input, moving to a random parent at each layer. At every node j on the path it opens the parents’ activations and the weights w_{ij} from C_M and checks $a_j = \phi(\sum_{i \in G_j} w_{ij} a_i)$, where G_j is the set of parents of j and ϕ the activation. A path opens one weight row per layer, so the proof and the prover’s work stay small.

The single-neuron attack. Our adversary runs the correct model on x , picks one neuron j , and replaces a_j by a value that makes the output wrong (untargeted) or equal to a class of its choice (targeted). To find j , it ranks neurons by the gradient of the decision margin and bisects for the smallest edit that changes the output. It then recomputes all later layers *with the true weights*. Every later node is consistent with its parents, so j is the only *inconsistent node* whose check fails. The verifier notices the attack only if its path visits j , which happens with probability $1/N$ in a layer of width N , or $1 - (1 - 1/N)^s$ with s paths.

3.3 Why sampling does not fix it

Anchuri et al. suggest sampling more paths, or sampling where it matters instead of uniformly. Neither helps against an adversary

who knows the rule. Fix a layer of width N and let U be the set of its neurons whose change can flip the output. If the sampler ignores the claimed values, the probabilities with which it opens each neuron are fixed before the adversary acts and sum to at most s over U , so the adversary can tamper with a neuron that is opened with probability at most $s/|U|$, on every network. A sampler that reads the values could do better on a particular network, but not on every one. We call a sampler *value-aware* if it chooses each neuron to open from the input, the claimed values, the weights of later layers, each neuron’s natural range (the values it takes on honest inputs) and the rows it has already opened.

PROPOSITION 3.1 (NO SAMPLER BEATS UNIFORM ON EVERY NETWORK). *For every value-aware sampler that opens at most s neurons of a hidden layer of width N and accepts every network’s honest trace, there is a network on which the probability that it catches a tampered neuron, kept within its natural range, is at most s/N .*

Proof. Let hidden neuron i copy input bit $\sigma(i)$ for a permutation σ , let natural inputs have at most one zero, and let the output be 1 exactly when all hidden neurons are 1. The adversary sets the neuron that should be 0 to 1, which flips the output and stays in the range $\{0, 1\}$. The forged trace, later weights and ranges do not depend on σ , and the forged trace is honest for the network in which that neuron copies another bit, so the sampler rejects only if it opens the tampered row. Every other row it opens shows a neuron that copies a 1, so for a random σ the tampered neuron stays uniform among the unopened neurons and is opened with probability at most s/N ; some σ does no better. $\square$

Opening s distinct neurons uniformly reaches s/N on every dense network, so nothing beats uniform sampling in the worst case, and catching one tampered neuron with constant probability needs $s = \Omega(N)$ openings per layer, the cost the protocol was designed to avoid. The *contribution-weighted sampler*, which steps to parent i with probability proportional to $|w_{ij} a_i|$, never visits a neuron whose claimed activation is zero, so zeroing out neurons can change the output undetected. A floor δ , sampling by $|w_{ij}|(|a_i| + \delta)$, leaves detection low (Section 4.2).

3.4 Our defence: checking every layer

Since no sampler has a better worst-case guarantee than uniform sampling, our verifier checks every neuron of every layer.

Exact integers. The checks compare exact values, but floating-point results differ between devices and summation orders. The model that is committed and checked is therefore an int8 version of M , obtained by integer-only quantisation [22], as many models are served; proof systems and TEE-based checks for inference likewise need integer or fixed-point models [9, 12, 17, 26, 32, 34]; Section 4.1 measures its accuracy. Each weight layer ℓ computes $Z_\ell = W_\ell X_\ell + b_\ell$, where W_ℓ is int8 with one scale per output channel, X_ℓ is int8 with one scale per tensor, and BatchNorm is folded into W_ℓ and b_ℓ ; convolutions become matrix products through im2col. These integer products can still use the GPU’s fast float32 units: float32 represents every integer of magnitude at most 2^{24} exactly and a product of int8 values is at most 2^{14} in magnitude, so the prover sums at most 1,024 of them at a time; like Slalom [34], it thus runs exact integer arithmetic in floating point, and its results are bit-identical to the verifier’s. Exactness also removes the floating-point

tolerance of the path protocol, which leaves an attacker room to hide small changes.

Three statements. With the output y , the prover sends the pre-activations Z_ℓ of every weight layer, which we call the *claims*. Write $A_\ell = [W_\ell | b_\ell]$. Between two weight layers the network applies only operations without weights, such as softmax and the attention products $QK^\top$ and PV . The answer y is the last weight layer’s pre-activations (the logits), so it is itself a claim, and it is correct if every claim is. Every claim equals the value that M computes on x if and only if three statements hold for every weight layer ℓ : (S1) its input X_ℓ follows from x and the previous claims through these operations, for example $X_\ell = \text{requant}(\text{ReLU}(Z_{\ell-1}))$ in an MLP; (S2) its claims are the product of its weights and its input, $Z_\ell = A_\ell[X_\ell; \mathbf{1}]$; and (S3) A_ℓ is the committed matrix.

Checking (S1): recomputation. The verifier computes every operation without weights itself, from x and the claims. Each input X_ℓ is the verifier’s own computation, not a value sent by the prover, and the checks are chained: a prover cannot pass one layer’s check with one value and feed another into the next. This is also why non-linear layers such as ReLU and softmax need no proof here.

Checking (S2): Freivalds’ check. Let A_ℓ have N rows (one per output neuron) and row length k , and let T be the number of columns of X_ℓ (pixels or tokens); we drop the subscript ℓ from N, k and T when the layer is clear. Recomputing $A_\ell[X_\ell; \mathbf{1}]$ is essentially performing the inference. Instead the verifier draws r independent random vectors $\chi \in \mathbb{F}_p^N$ and, for each, checks one random combination of the layer’s rows,

$$\chi^\top Z_\ell = \underbrace{(\chi^\top A_\ell)}_{u_\ell} [X_\ell; \mathbf{1}], \quad (1)$$

where we call u_ℓ the *folded row*. Let $D = Z_\ell - A_\ell[X_\ell; \mathbf{1}]$ be the error of the claims: its entry D_{im} is how far the claim of neuron i at position m (a pixel or token) is from the true value, so a correct layer has $D = 0$. The two sides of Eq. (1) differ by $\chi^\top D$, so the check passes only if $\chi^\top D = 0$. If some $D_{im} \neq 0$, entry m of $\chi^\top D$ is $\sum_{i'} \chi_{i'} D_{i'm} = \chi_i D_{im} + c$, where c does not involve χ_i . Whatever the other entries of χ are, this is zero for only one of the p values χ_i can take ($\chi_i = -c/D_{im}$), so a wrong layer passes with probability at most $1/p$; it must pass the check for each of the r vectors, so at most p^{-r} overall. This holds however many values are wrong and wherever they are: a single tampered neuron is caught as reliably as an entirely wrong layer. Given u_ℓ (cost $O(rNk)$), the two sides cost $O(r(N+k)T)$, against $O(NkT)$ for recomputing the layer.

We work in the BabyBear field $\mathbb{F}_p$, $p = 15 \cdot 2^{27} + 1 \approx 2^{31}$. Integers map into $\mathbb{F}_p$ without wrap-around as long as they are small, and two checks ensure this. The verifier range-checks every claim ($|z| < 2^{29}$), and setup checks that every honest pre-activation is in the same range: $128 \sum_c |w_{ic}| + |b_i| < 2^{29}$ for every row i of a product, whose inputs are int8 and so at most 128 in magnitude (for an embedding, $\max_c |w_{ic}| + |b_i|$). A wrong claim and the correct value then differ by less than $2^{30} < p$, so they remain distinct in $\mathbb{F}_p$.

Checking (S3): committed weights. In K and K_{pre} the verifier holds the weights and computes u_ℓ on its own, so (S3) holds by construction. In setting C it cannot, so the prover sends u_ℓ , and the verifier must check that $u_\ell = \chi^\top A_\ell$ for the committed A_ℓ . Opening a few entries of u_ℓ would miss one changed entry, as a path misses

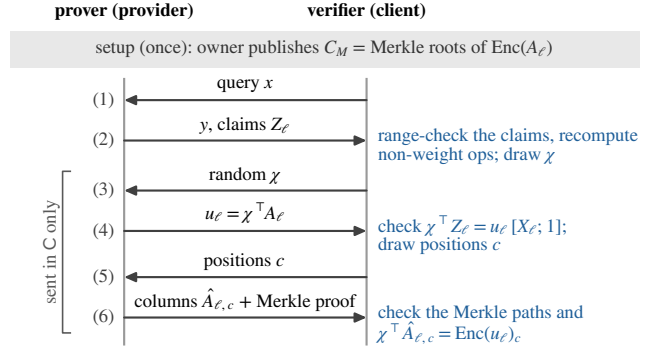


Figure 2: One query in setting C: each value is fixed before the randomness that tests it

one neuron, so we commit to an *encoded* copy of the weights, as in the Ligero and Brakedown commitments [1, 19]. At setup every row of A_ℓ is Reed–Solomon encoded [31]: the polynomial whose coefficients are the row is evaluated at $n_\ell = 4\bar{k}$ points, where $\bar{k} \geq k$ is a power of two (Section 3.6 relaxes this), giving $\hat{A}_\ell$. The commitment C_M holds one Merkle root [28] per weight matrix, over the columns of $\hat{A}_\ell$. Encoding is linear, so $\chi^\top \hat{A}_\ell = \text{Enc}(\chi^\top A_\ell)$. The verifier encodes the claimed u_ℓ itself, opens t random columns c with their Merkle paths, and checks whether $\chi^\top \hat{A}_{\ell,c} = \text{Enc}(u_\ell)_c$ holds. If u_ℓ is wrong, $\text{Enc}(u_\ell)$ and $\text{Enc}(\chi^\top A_\ell)$ are distinct codewords whose difference is a nonzero polynomial of degree below k , so they agree on at most $k-1$ of the n_ℓ positions, and t distinct columns all miss the difference with probability at most $((k-1)/n_\ell)^t < 4^{-t}$.

3.5 Protocol and soundness

Setup happens once per model: the owner encodes and commits the weights and publishes C_M . Figure 2 shows one query of the basic protocol, with r vectors χ and t columns per layer; besides recomputing the operations without weights (S1) and the checks of (S2) and (S3), the verifier range-checks the claims.

THEOREM 3.2 (SOUNDNESS). *If C_M is honest and the prover cannot find a hash collision, then for any claimed $y \neq M(x)$ the verifier accepts with probability at most $\epsilon = \sum_\ell [p^{-r} + (\frac{k_\ell-1}{n_\ell})^t]$.*

Proof. Some claim is wrong, since y is one; let ℓ be the first layer with a wrong claim. Its input X_ℓ is correct, because the verifier computed it from x and correct earlier claims (S1), and the range check keeps the wrong claim distinct from the true value in $\mathbb{F}_p$. Hence $Z_\ell \neq A_\ell[X_\ell; \mathbf{1}]$ for the committed A_ℓ , and the prover passes only by fooling a random check: with the true u_ℓ , Eq. (1) fails except with probability p^{-r} (S2); with another u_ℓ , a column check fails except with probability $((k_\ell-1)/n_\ell)^t$ (S3), since the opened columns are the committed ones unless the prover finds a hash collision. A union bound over the layers and these two cases finishes the proof. $\square$

Parameters. For a target λ and L weight layers we set $\beta = \lambda + \log_2(2L)$, $r = \lceil \beta / \log_2 p \rceil$ and $t = \lceil \beta / 2 \rceil$; since $n_\ell \geq 4k_\ell$, each of the $2L$ terms of ϵ is at most $2^{-\beta}$, so $\epsilon \leq 2^{-\lambda}$ (for VGG-16 at $\lambda=128$, $r=5$ and $t=67$). Interactively, each challenge is expanded

with SHAKE-256 from a fresh 256-bit random key; deriving the keys from a hash of the transcript (Fiat-Shamir [14]) makes the protocol non-interactive. Once the claims are fixed, only the first wrong layer’s χ can make them pass, and once u_ℓ is fixed only that layer’s columns can. The interactive protocol is thus round-by-round sound [6], hence state-restoration sound [6, 21], so by [4] a cheating prover that makes Q hash (random-oracle) queries, for instance rehashing altered claims in search of favourable challenges, succeeds with probability at most $Q 2^{-\beta}$ plus that of finding a collision, which 64 extra bits in β keep below $2^{-\lambda}$ for $Q \leq 2^{64}$.

Known weights. In K the verifier computes u_ℓ itself, so messages (3)–(6) are unnecessary and omitted. In K_{pre} it computes the secret χ and u_ℓ once per model, so the proof is the claims alone. The prover learns each verdict and could repeat an accepted forgery, but until one is accepted the verdicts do not depend on χ , so a false acceptance occurs among Q queries that reuse χ with probability at most $Q \sum_\ell p^{-r}$.

3.6 Costs and optimisations

3.6.1 Costs. The prover runs the forward pass, computes u_ℓ and opens t columns, which it recomputes from W_ℓ ; its extra work, $O((r+t) \sum_\ell N_\ell k_\ell)$, does not depend on the number of pixels or tokens. The verifier recomputes the operations without weights, $O(N_\ell T_\ell)$ per layer plus attention’s $O(T^2 d)$ for language models; checks Freivalds’ equation, $O(r(N_\ell + k_\ell)T_\ell)$; and checks the columns, encoding the folded rows ($O(rn_\ell \log n_\ell)$), combining the opened columns ($O(rtN_\ell)$) and hashing the Merkle paths. No term grows with $N_\ell k_\ell T_\ell$, the cost of the layer’s matrix product, and only attention, which inference also computes, grows faster than linearly in T . With 4-byte claims and field elements and 32-byte hashes, a proof in setting C has

$$|\pi| = 4 \sum_\ell N_\ell T_\ell + 4r \sum_\ell k_\ell + 4t \sum_\ell N_\ell + 32 \sum_\ell h_\ell \quad (2)$$

bytes: the claims, which grow with the input, and the folded rows, the t opened columns of N_ℓ entries and the $h_\ell \leq t \log_2 n_\ell$ hashes of each layer’s Merkle multiproof, which do not. For Llama-2-7B at 64 tokens these are 349, 23, 385 and 4 MB of the basic protocol’s 762 MB proof. In K and K_{pre} the proof is the claims alone.

3.6.2 Optimisations. The parameters above keep the soundness argument short but leave the proof larger than it needs to be. We call the protocol so far the *basic* protocol, and the one with the four changes below and the engineered verifier described after them the *optimised* protocol, which Section 4 measures and Section 4.5 compares with the basic one. The changes alter only how the weights are committed and checked, which positions of the last block are proved for a next-token output, and how messages are sent. Theorem 3.2 still holds, with one term per encoded matrix j (a looked-up embedding table adds none): p^{-r} , plus the chance that its t_j distinct opened columns all fall among the at most $m_j - 1$ of its n_j positions where two distinct codewords of message length m_j agree.

Fewer columns. The basic protocol opens t columns of every matrix, each under its own Merkle tree. A column exposes a wrong value unless it lands where the honest and the claimed codeword agree, so a codeword long relative to its message needs fewer columns. The optimised protocol opens in each tree just enough

columns to keep every term below $2^{-\beta}$, and matrices with codewords of equal length can share one tree and multiproof. A public *planning rule* picks, of six candidate plans of codeword lengths and layouts, the one sending the fewest bytes per query besides the claims, with at most twice the basic protocol’s encoded entries at setup (or 2^{34} if more); the plan follows from public shapes and is part of C_M . LeNet-5’s matrices then share one tree of length 2^{18} and open 15 columns instead of 66 in each of five trees.

Transposed layout. A layer’s opening costs $rk + tN$ field elements, too many for a wide layer such as GPT-2’s output layer ($N=50,257$). For such layers the owner instead commits the encoded rows $\hat{A}'$ of $A^\top$, and the verifier combines positions rather than neurons: it draws r vectors $\chi' \in \mathbb{F}_p^T$ (the T unit vectors when $T \leq r$), computes $z = Z\chi'$ and $w = [X; \mathbf{1}]\chi'$ itself for each, and checks $\text{Enc}(z)_c = (w^\top \hat{A}')_c$ on t opened columns of k entries. Since $w^\top \hat{A}' = \text{Enc}(Aw)$, a wrong claim passes only if all r products $(Z - A[X; \mathbf{1}])\chi'$ vanish or all t columns fall where two distinct codewords agree (here m_j counts the stacked rows). No folded row is sent, and the opening shrinks to tk field elements; projections of one input may share one such matrix.

Decoders. An embedding table without a bias is only looked up, so, as in CommitLLM [7], it is committed as a Merkle tree over its rows and the prover opens the rows it used; a wrong row would need a hash collision. A next-token output depends only on the last position, so the last block is computed there alone, except for the keys and values that attention needs at every position.

Compact encoding. An int8 model’s claims need far fewer than the 32 bits we count for each. Each layer’s claims are bit-packed at a width chosen for it, the few values that do not fit sent separately (frame-of-reference and Rice coding), and field elements in 31 bits; a claim then takes 16–19 bits. The verifier decodes before checking; Fiat-Shamir absorbs the encoded bytes. In K and K_{pre} , which open no columns, only this change and the previous one apply, the verifier reading the embedding rows itself.

Verifier engineering. Without changing a proof byte, the optimised verifier evaluates $\text{Enc}(u_\ell)$ only at the opened positions, batches its exact products, computes attention in int32 (as the released prover also does) and streams the claims to a GPU.

4 Results

4.1 Experimental setup

Models. LeNet-5 (MNIST) and VGG-11, VGG-16 and ResNet-18 (CIFAR-10) are the CNN architectures that zkCNN, ZKML and Bionetta report on; Table 1 compares only the published versions that match ours (ZKML’s ResNet-18, with 281K parameters, does not). A 224×224 dog-vs-cat ResNet-18 trained on ImageNet classes stands in for the unreleased model of [2], a dog-vs-squirrel one for its substitute $\tilde{M}$, and the MNIST MLP (784–512–256–10) is the network attacked in Section 4.2. We trained all of them, and int8 quantisation changed accuracy by at most 0.4 points. The language models have the shapes of GPT-2, OPT-125M to OPT-13B, Llama-2-7B/13B and Qwen3-4B, built in full up to 13B and 2,048 tokens, with random int8 weights; the costs depend only on the shapes, except the compact encoding’s size, which may differ for real weights.

Real weights. To check that the integer graph is a usable language model, we load the real OPT-125M, OPT-1.3B and OPT-6.7B

into it (with one extra bias on the LM head, from folding the final LayerNorm). On 4,064 tokens of WikiText-2 their perplexity is 73.8, 37.4 and 36.0, against 64.7, 34.9 and 27.1 in fp32, slightly optimistic because two public constants (the LayerNorm output gain, 4, 3 and 2, and the calibration percentile) were chosen on this text.

Hardware and baselines. Our implementation is in Python and PyTorch, with SHA-256 Merkle trees. The prover is one NVIDIA L40S (48 GB) and the verifier 8 threads of an AMD EPYC 9334; times are medians over a run’s queries. Of the 30–70B shapes only 1- and 2-block builds fit, from which we extrapolate proof sizes (within 2.83% wherever a full build exists) but not times. Published costs are as reported on their own hardware, each checked against its source; for the path protocol we compute exact detection probabilities and bytes on our graphs and take its time per path from Anchuri et al. The basic protocol was measured at commit 9401431 (a rerun on an RTX 2080 Ti reproduced all 1,978 hardware-independent values), the optimised one with the released code (c8be5eb and three successors that change no proof byte), which reproduces the basic proofs byte for byte.

4.2 Breaking the path protocol

Reproduction. Our implementation behaves as Anchuri et al. describe. On MNIST versions of four of their substitute-model settings, honest provers were accepted in all 3,000 runs of each, and substitute models were caught in 99.8–100% of runs, even an 8-bit quantised copy that agrees with M on all 150 evaluated queries. Without the 10^{-4} tolerance that Anchuri et al. suggest, an honest prover is accepted in 21–37% of runs, because a layer-wide matrix product and a single-neuron dot product round differently.

Attacks on MNIST. We compute each attack’s exact probability of passing one path by dynamic programming over the paths. The forgeries of [2] (its substitute model, Appendices E and F and Section 7.2) corrupt hundreds of the MLP’s 778 trace nodes and are always caught, while ours leave 1–14 inconsistent nodes. One tampered neuron is accepted with the predicted probability $1 - 1/N$ (99.6–99.8%). Triggered by a 3×3 bright patch, the backdoor returns the honest output on every clean query and succeeds on every triggered query not already in the target class. It passed 99.96% of 2,300 end-to-end runs, within sampling error of the predicted 99.8%. Keeping each tampered value below its neuron’s 95th percentile takes about five neurons and still passes 99% of the time under uniform sampling. With 250 paths, the single-neuron attack is accepted 61% of the time.

Other samplers. Against an adversary who knows the rule, uniform sampling detects 0.20% of attacks, a static importance score (outgoing weight magnitude) 0.11% and gradient saliency 0.016%. The contribution-weighted sampler detects nothing: zeroing about 14 of the 512 neurons, their most common value (65% of inputs), flips the prediction, and end-to-end runs accepted 2,000 of 2,000 attacks. A floor $\delta = 1$ (sampling by $|w_{ij}|(|a_i| + \delta)$) raises detection to 2% (10× uniform); other floors give 0.9–6×.

Larger models. On every CNN and every test query, changing one penultimate neuron suffices to flip the prediction. One path catches it with probability 1/84 (LeNet-5) to 1/512 (VGG, ResNet), and a change to the least-visited neuron with probability as low as

Optimised at 128 bits sends less than paths at 40 bits

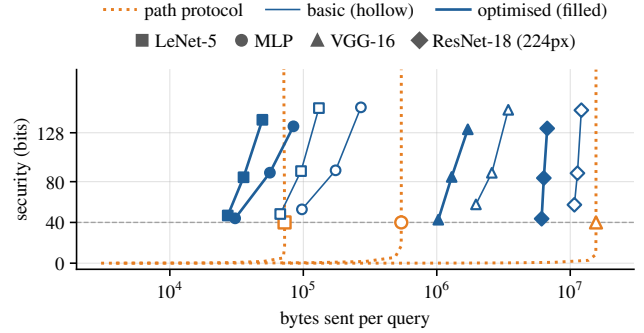


Figure 3: Security against bytes sent: our proofs buy far more security per byte than the paths

1/28 million. This includes the dog-vs-cat ResNet-18, a binary classifier, for which Anchuri et al. argue that their analysis extends to full soundness because the adversary can only flip the output. Reaching 2^{-40} takes 2,316–14,182 paths, which open the whole model and trace (Figure 3). On Llama-2-7B one path detects the attack with probability 1/11,008, and 2^{-40} needs about 305,000 paths (≈ 13.7 GB with the fp16 weights of [2], at 64 tokens).

A real LLM. On the real OPT-6.7B (fp16), the same search over FFN neurons of layers 8, 16, 24 and 31 at the last position, also ranked by their direct effect on the logits, changed the next token with one neuron for all 40 prompts. A fixed-target backdoor is more constrained: because of the final LayerNorm, a last-layer neuron can force only its *saturation token*, so the 16,384 neurons reach 6,187 of the 50,272 tokens. The backdoor succeeds for “back” (40/40 triggered prompts, clean prompts unchanged) and fails for “hacked” (0/40). Its values exceed the neurons’ natural range, which the path protocol does not check; one path detects it with probability 1/16,384, and 2^{-40} needs 454,248 paths.

4.3 Soundness in practice

Outcomes. In the optimised runs the verifier accepted all 3,792 honest queries and rejected all 2,512 attacks. The 1,848 image-model attacks, at $\lambda=40$ with committed weights, were one changed value, one changed penultimate neuron, one changed logit, 1% of the weights perturbed, the substitute $\tilde{M}$, and two forgeries that each pass every check but one. Every language-model run added +1 to one middle-layer pre-activation (106 attacks, 20 of them on 1- and 2-block builds of the 30–70B shapes), and for five models from OPT-1.3B to Llama-2-13B we also ran the image-model attacks that apply to them. The basic protocol’s run accepted all 8,709 honest queries and rejected all 2,895 attacks (804 on language models).

Each check is needed. Freivalds’ check rejected 1,601 attacks and the column check against the code 853, including every forged folded row and every tampered claim of a layer in the transposed layout; the Merkle check rejected all 56 forged columns (a range check caught the remaining 2).

Security per byte. Figure 3 plots security against the bytes sent per query for the image models, each with its own marker, orange

Table 2: Image models: proving takes milliseconds

Model	Params	acc.	committed (C)			K _{pre}	
			prove↓	verify↓	proof↓	verify↓	proof↓
MLP (MNIST)	536K	97.7%	3.8 ms	1.7 ms	84.3 kB	0.7 ms	1.9 kB
LeNet-5 (MNIST)	62K	99.3%	5.3 ms	2.5 ms	49.4 kB	1.6 ms	14.2 kB
VGG-11 (CIFAR)	9.8M	91.1%	12.8 ms	11.7 ms	1.0 MB	6.5 ms	334 kB
VGG-16 (CIFAR)	15.2M	93.0%	17.7 ms	17.2 ms	1.7 MB	10.1 ms	610 kB
ResNet-18 (CIFAR)	11.2M	94.2%	20.0 ms	24.9 ms	2.3 MB	16.4 ms	1.3 MB
ResNet-18 (224px) [†]	11.2M	95.0%	22.9 ms	64.8 ms	6.7 MB	59.2 ms	5.7 MB

[†]Our ImageNet-trained counterpart of the classifier of [2].

Table 3: Language models: seconds to prove, but proofs grow with the prompt (–: not measured)

Model	Params	committed weights (C)				K _{pre}	
		prove↓	verify↓	GPU verify↓	proof↓	verify↓	proof↓
64-token prompts							
GPT-2	124M	76.1 ms	76.9 ms	79.9 ms	14.8 MB	61.8 ms	10.9 MB
Qwen3-4B	4.0B	1.2 s	712 ms	–	223 MB	692 ms	160 MB
Llama-2-7B	6.7B	1.8 s	1.0 s	431 ms	325 MB	698 ms	196 MB
Llama-2-13B	13.0B	2.7 s	1.5 s	–	469 MB	1.1 s	313 MB
2,048-token prompts							
OPT-125M	125M	396 ms	4.8 s	318 ms	348 MB	4.6 s	341 MB
OPT-1.3B	1.3B	3.8 s	27.4 s	2.6 s	2.1 GB	27.5 s	2.0 GB
OPT-6.7B	6.7B	7.7 s	46.5 s	3.5 s	5.5 GB	50.1 s	5.4 GB
Llama-2-7B	6.7B	–	–	–	6.6 GB	58.2 s	6.4 GB
Llama-2-13B	13.0B	17.6 s	–	–	10.5 GB	84.5 s	10.3 GB

for its paths and blue for our protocols (the 224-pixel ResNet’s paths, which coincide with VGG-16’s, are not drawn). Security, in bits, is $-\log_2$ of the probability that a wrong output is accepted: for the paths, that the single-neuron attack is missed, and for ours, the proven bound at $\lambda=40, 80$ and 128. The paths reach 40 bits only by opening the whole model and trace (the vertical lines). The optimised protocol lands close to its target (43–47 bits at $\lambda=40$ and 131–141 at $\lambda=128$) and, on every model, reaches 2^{-128} with fewer bytes than the paths need for 2^{-40} . The basic protocol overshoots its target because r and t are rounded up (48–58 and 150–153 bits), and at $\lambda=40$ it sends 1.1–8.3× fewer bytes than the paths need for 2^{-40} .

4.4 Cost

Image models. Table 2 gives the cost of one query to the optimised protocol at $\lambda=128$. With committed weights the prover needs 3.8–22.9 ms per query and the verifier 1.7–64.8 ms. The proof, 49.4 kB–6.7 MB, is 1.3–9.3× smaller than the int8 weights.

Language models. Table 3 proves the next-token logits of one prompt at $\lambda=128$ with random int8 weights; its GPU verify column runs our verifier on the GPU. With a 64-token prompt GPT-2 is proved in 76.1 ms and Llama-2-7B in 1.8 s, and Llama-2-7B’s 325 MB proof is about 21× smaller than its int8 weights. At 2,048 tokens the same-sized OPT-6.7B is proved in 7.7 s, but its proof grows to 5.5 GB and CPU verification to 46.5 s. The proof grows because the claims grow with the prompt (the K_{pre} proof, which holds only claims, is almost as large), and the verifier because it recomputes attention over all token pairs; in setting C a GPU verifier

is 11–15× faster there and 1.0–2.4× at 64 tokens. Prompts whose claims are all sent before the challenge can share one set of challenges and columns; in batches of eight 64-token prompts, Llama-2-7B’s prover then needs 0.39 s and its proof 219 MB per prompt. For Llama-2-70B the extrapolated proof at 64 tokens is 1.7 GB, 41× smaller than its int8 weights.

Figure 4 places these runs among the published ones, with our verifier on 8 CPU threads and the published systems (and the path of [2]) as reported on their own hardware. Part of the prover gap is our GPU, since all the zkSNARKs except zkLLM ran on CPUs.

4.5 Effect of the optimisations

We ran both protocols on the same machines (setting C, interactive, $\lambda=128$) on LeNet-5, VGG-16, ResNet-18 (CIFAR), GPT-2 (64 and 512 tokens) and Llama-2-7B (1 and 64 tokens). On average (geometric mean) the optimised protocol sends 61% fewer bytes (2.0–4.2× fewer), verifies in 77% less time (2.3–10.9× faster) and proves in 28% less time. Most of the verifier gain comes from verifier engineering, which also speeds up the basic proofs (468 to 190 ms for GPT-2 at 64 tokens), and the prover is faster on every model but LeNet-5, where encoding the claims costs more than it saves. At 2,048 tokens, where the claims dominate, the optimised protocol shrinks Llama-2-7B’s proof from 11.6 to 6.6 GB. With the basic proof format, the prover of Llama-2-7B at 2,048 tokens takes 1.9× the time of the int8 forward pass. The price is that the one-time setup is longer (~ 115 s instead of ~ 10 s for GPT-2, ~ 8.6 instead of 4.5 min for Llama-2-7B).

4.6 Comparison with published systems

In Table 1, which compares the optimised protocol with published systems on matched models, our prover wins in every row, while the verifier and the proof win in some rows and lose in others. Our verifier runs on 8 CPU threads at $\lambda=128$, except against Maverick (note *d*) and in the 2,048-token rows, where we use our GPU verifier, since zkLLM also runs on a GPU; the prover times and proof sizes there are those of Table 3, as the choice of verifier does not change the proof. For the language models ours prove the next-token logits. Most systems we compare with produce non-interactive proofs, so against them we make ours non-interactive with Fiat–Shamir (Section 3.5), which costs us extra columns; zkLLM is itself interactive, so against it we run ours interactively. We use committed weights except against Maverick, whose client holds the weights, so we compare it with K_{pre}; unlike K_{pre}, it draws a fresh public challenge per query.

Prover. Our prover is faster than every system in Table 1. It is faster than every zkSNARK prover, by 58× (zkLLM, OPT-1.3B) to 32,000× (ZKML, VGG-16) and 3.0× faster than Maverick.

Verifier. Our verifier is faster in most rows of Table 1; it is slower only than ZKML’s on VGG-16 (2.2×) and zkLLM’s on OPT-1.3B at 2,048 tokens (2.9×). Both have provers that take minutes per query. Against Maverick, we are better on all three costs, with or without the compact encoding (note *d*). With the encoding our proof is smaller (20.3 MB, 1.8× smaller than Maverick’s), and our verifier, at 151.6 ms, is still faster than Maverick’s 260.2 ms.

Proof. Our proof is larger in most rows, since it carries every layer’s claims and opened columns: by 3.6–180× against the other

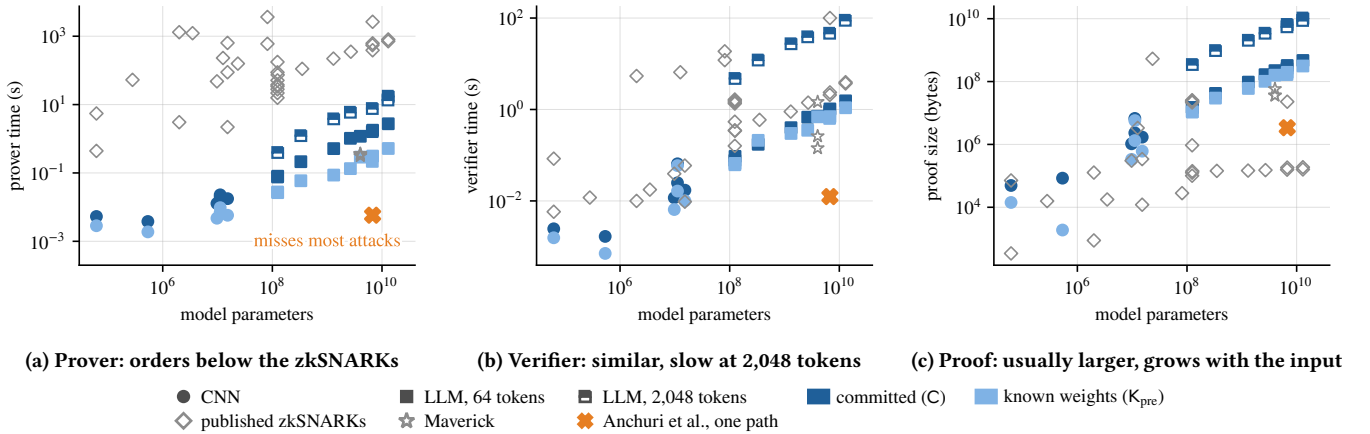


Figure 4: Cost against model size: ours (blue) proves far faster than published systems (grey), but usually sends more

short-input proofs and by 2,500–14,000 $\times$ against zkLLM, whose proof stays small even at 2,048 tokens. It is smaller in three rows, and in each of them we win on all three costs.

Summary. Our prover is the fastest of every fully sound system we compare with, by orders of magnitude against the zkSNARKs: it proves a query in milliseconds for the image models and in seconds for the language models. Our verifier is faster than every other fully sound system’s except ZKML’s and zkLLM’s, whose provers take minutes per query, too slow to prove every answer. Against zkCNN on LeNet-5, DeepProve on GPT-2 at 64 tokens and Maverick we are better on all three costs. We get there with only hashing and Freivalds’ check, without a SNARK or a trusted setup; the one cost we pay is a larger proof, which grows with the input.

5 Discussion

Implications. Anchuri et al.’s path protocol still catches a provider that substitutes a cheaper model, but one that wants to change a particular answer passes it almost surely, so it should not be relied on where a wrong answer is costly. Our protocol catches such a provider except with probability $2^{-\lambda}$, at a prover cost of milliseconds to seconds per query and without a zkSNARK or a trusted setup, which makes checking every answer practical. MLPs and CNNs are verified in milliseconds on an ordinary CPU; for language models with long prompts the verifier needs a GPU and a fast link, so it fits a gateway or an auditor rather than an end device.

Limitations. (i) The proof grows with the activations; at 2,048 tokens it is mostly claims and gigabytes for 7B models, against zkLLM’s 157–183 kB. (ii) The verifier’s cost grows with the square of the prompt length: at 2,048 tokens the CPU verifier needs up to 84.5 s, and the GPU verifier is up to $2.9\times$ slower than zkLLM’s. (iii) The provider must serve the integer model, as proof systems and TEE-based checks for inference also need integer or fixed-point models [9, 12, 17, 26, 32, 34], at some cost in accuracy (Section 4.1). (iv) Like the path protocol, we assume the weight commitment is computed honestly; longer codewords make it up to $13\times$ slower to compute for GPT-2. Accepting a commitment

from an untrusted party would require a proximity argument as in Ligerio. (v) The opened columns reveal random linear combinations of each weight row, so the protocol is not zero-knowledge.

Future work. The proof is mostly claims, which the verifier needs in full to recompute the operations without weights; proving those operations with a sum-check argument, as zkLLM does, would let it check them at a few random points instead, at the cost of more prover work. Checking the attention products instead of recomputing them, with Freivalds’ check on claimed scores or a sum-check argument as in zkLLM [32], could make the verifier much faster at long prompts, at some cost in proof size. A native verifier and a KV cache would also help.

6 Conclusion

We showed that the path protocol of Anchuri et al. [2] cannot stop a provider that wants to change an answer: one that tampers with a single neuron and computes everything else honestly passes its check almost always, on image classifiers and on the real OPT-6.7B, and no sampling rule, however it chooses where to look, closes this gap on every network.

We close the gap by checking every layer instead of a sample. It needs only hashing, with no SNARK, trusted hardware or trusted setup, and it is practical: its prover is the fastest of all the fully sound systems we compared, orders of magnitude faster than the zkSNARK provers; its verifier is faster than every other fully sound system’s except ZKML’s and zkLLM’s, whose provers take minutes per query; and against zkCNN on LeNet-5, DeepProve on GPT-2 at 64 tokens and Maverick it is better on all three costs. Its remaining cost is a proof that grows with the input, which makes long prompts its weak point and the natural target for future work.

7 Contributions

We are close friends so we did most of this work sitting together. Edo Kadosh focused somewhat more on the optimisations, Ofek Lavi on the attack and the sampling proofs, and Erel Barzilay on the basic defence, but the vast majority of the work was joint. We used Claude to help write the code and correct our grammar.

References

- [1] Scott Ames, Carmit Hazay, Yuval Ishai, and Muthuramakrishnan Venkitasubramaniam. 2017. Ligerio: Lightweight Sublinear Arguments Without a Trusted Setup. In *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2087–2104. doi:10.1145/3133956.3134104
- [2] Pranay Anchuri, Matteo Campanelli, Paul Cesaretti, Rosario Gennaro, Tushar M. Jois, Hasan S. Kayman, and Tugce Ozdemir. 2026. Towards Verifiable AI with Lightweight Cryptographic Proofs of Inference. In *2026 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*. IEEE, 955–977. doi:10.1109/SaTML68715.2026.00058
- [3] Arasu Arun, Adam St. Arnaud, Alexey Titov, Brian Wilcox, Viktor Kolobaric, Marc Brinkmann, Oguzhan Ersoy, Ben Fielding, and Joseph Bonneau. 2025. Verde: Verification via Refereed Delegation for Machine Learning Programs. arXiv:2502.19405.
- [4] Eli Ben-Sasson, Alessandro Chiesa, and Nicholas Spooner. 2016. Interactive Oracle Proofs. In *Theory of Cryptography (TCC 2016-B) (LNCS, Vol. 9986)*. Springer, 31–60.
- [5] Wyatt Benno, Alberto Centelles, Antoine Douchet, and Khalil Gibran. 2026. Jolt Atlas: Verifiable Inference via Lookup Arguments in Zero Knowledge. arXiv:2602.17452.
- [6] Ran Canetti, Yilei Chen, Justin Holmgren, Alex Lombardi, Guy N. Rothblum, Ron D. Rothblum, and Daniel Wichs. 2019. Fiat-Shamir: From Practice to Theory. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC)*. ACM, 1082–1090.
- [7] Federico Carrone, Diego Kingston, Manuel Puebla, and Mauro Toscano. 2026. CommitLLM: A Commit-and-Audit Protocol for Open-Weight LLM Inference. Preprint, <https://github.com/lambdaclass/CommitLLM>. Paper at paper/main.pdf, commit 25541e83.
- [8] Bing-Jyue Chen, Lilia Tang, and Daniel Kang. 2025. ZKTorch: Compiling ML Inference to Zero-Knowledge Proofs via Parallel Proof Accumulation. arXiv:2507.07031.
- [9] Bing-Jyue Chen, Suppakit Waiwitikhit, Ion Stoica, and Daniel Kang. 2024. ZKML: An Optimizing System for ML Inference in Zero-Knowledge Proofs. In *Proceedings of the Nineteenth European Conference on Computer Systems (EuroSys)*. ACM, 560–574. doi:10.1145/3627703.3650088
- [10] Ziqun Chen, Ming Wu, Michael Heinrich, Jason Zeng, Huiying Lan, Tianwei Zhang, and Rui Tan. 2026. Communication-Efficient Verifiable Attention for LLM Inference. arXiv:2606.16352. Presents VeriAttn.
- [11] KD Conway, Cathie So, Xiaohang Yu, and Kartini Wong. 2024. opML: Optimistic Machine Learning on Blockchain. arXiv:2401.17555.
- [12] Boyuan Feng, Lianke Qin, Zhenfei Zhang, Yufei Ding, and Shumo Chu. 2021. ZEN: An Optimizing Compiler for Verifiable, Zero-Knowledge Neural Network Inferences. Cryptology ePrint Archive, Paper 2021/087. <https://eprint.iacr.org/2021/087>
- [13] Boyuan Feng, Zheng Wang, Yuke Wang, Shu Yang, and Yufei Ding. 2024. ZENO: A Type-based Optimization Framework for Zero Knowledge Neural Network Inference. In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*. ACM, 450–464. doi:10.1145/3617232.3624852
- [14] Amos Fiat and Adi Shamir. 1986. How to Prove Yourself: Practical Solutions to Identification and Signature Problems. In *Advances in Cryptology – CRYPTO ’86 (LNCS, Vol. 263)*. Springer, 186–194.
- [15] Rusins Freivalds. 1977. Probabilistic Machines Can Use Less Running Time. In *Information Processing 77, Proceedings of the IFIP Congress*. North-Holland, 839–842.
- [16] Nicolas Gailly, Ismael Hishon-Rezaizadeh, Tianyi Liu, Nicholas Mainardi, Dimitrios Papadopoulos, Charalampos Papamanthou, Christodoulos Pappas, Shraavan Srinivasan, Zack Youell, and Yupeng Zhang. 2026. DeepProve: Verifiable End-to-End Large Language Model Inference. Cryptology ePrint Archive, Paper 2026/1112. To appear in ACM CCS 2026.
- [17] Zahra Ghodsi, Tianyu Gu, and Siddharth Garg. 2017. SafetyNets: Verifiable Execution of Deep Neural Networks on an Untrusted Cloud. In *Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30.
- [18] Shafi Goldwasser, Yael Tauman Kalai, and Guy N. Rothblum. 2008. Delegating Computation: Interactive Proofs for Muggles. In *Proceedings of the 40th Annual ACM Symposium on Theory of Computing (STOC)*. ACM, 113–122.
- [19] Alexander Golovnev, Jonathan Lee, Srinath Setty, Justin Thaler, and Riad S. Wahby. 2023. Brakedown: Linear-Time and Field-Agnostic SNARKs for R1CS. In *Advances in Cryptology – CRYPTO 2023*. Springer, 193–226. doi:10.1007/978-3-031-38545-2_7
- [20] Kanav Gupta, Jonathan Katz, and Ian Miers. 2026. Private and Verifiable Outsourcing of Open-Weight LLM Inference. Cryptology ePrint Archive, Paper 2026/1849.
- [21] Justin Holmgren. 2019. On Round-By-Round Soundness and State Restoration Attacks. Cryptology ePrint Archive, Paper 2019/1261.
- [22] Benoit Jacob, Skirmantas Kligys, Bo Chen, Menglong Zhu, Matthew Tang, Andrew Howard, Hartwig Adam, and Dmitry Kalenichenko. 2018. Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2704–2713.
- [23] Kyeongtae Lee, Byeongkyu Han, Jihye Kim, and Hyunok Oh. 2026. LAMP: Linear Verification of Matrix Multiplication via Proximity Testing. Cryptology ePrint Archive, Paper 2026/1571.
- [24] Seunghwa Lee, Hankyung Ko, Jihye Kim, and Hyunok Oh. 2024. vCNN: Verifiable Convolutional Neural Network Based on zk-SNARKs. *IEEE Transactions on Dependable and Secure Computing* 21, 4 (2024), 4254–4270. doi:10.1109/TDSC.2023.3348760
- [25] Youki Lim and Sam Yong. 2026. Seal, Then Sample: Sampled Layerwise Proofs for Verifiable LLM Inference from GPT-2 to 70B. arXiv:2609.27367.
- [26] Tianyi Liu, Xiang Xie, and Yupeng Zhang. 2021. zkCNN: Zero Knowledge Proofs for Convolutional Neural Network Predictions and Accuracy. In *Proceedings of the 2021 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2968–2985. doi:10.1145/3460120.3485379
- [27] Ben Merbaum, Mohammad Amin Raeisi, Wenhao Wang, Charalampos Papamanthou, Katerina Sotiraki, and Fan Zhang. 2026. Maverick: Private and Verifiable LLM Inference Made Practical via Matrix-Vector Multiplication Delegation. arXiv:2609.10264.
- [28] Ralph C. Merkle. 1987. A Digital Signature Based on a Conventional Encryption Function. In *Advances in Cryptology – CRYPTO ’87 (LNCS, Vol. 293)*. Springer, 369–378.
- [29] Jack Min Ong, Matthew Di Ferrante, Aaron Pazdera, Ryan Garner, Sami Jaghouar, Manveer Basra, Max Ryabinin, and Johannes Hagemann. 2025. TOPLOC: A Locality Sensitive Hashing Scheme for Trustless Verifiable Inference. In *Proceedings of the 42nd International Conference on Machine Learning (ICML) (PMLR, Vol. 267)*. 47196–47211.
- [30] Wenjie Qu, Yijun Sun, Xuanming Liu, Tao Lu, Yanpei Guo, Kai Chen, and Jiaheng Zhang. 2025. zkGPT: An Efficient Non-interactive Zero-knowledge Proof Framework for LLM Inference. In *34th USENIX Security Symposium*. USENIX Association, 2045–2063.
- [31] Irving S. Reed and Gustave Solomon. 1960. Polynomial Codes over Certain Finite Fields. *J. Soc. Indust. Appl. Math.* 8, 2 (1960), 300–304.
- [32] Haochen Sun, Jason Li, and Hongyang Zhang. 2024. zkLLM: Zero Knowledge Proofs for Large Language Models. In *Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 4405–4419. doi:10.1145/3658644.3670334
- [33] Yifan Sun, Yuhang Li, Yue Zhang, Yuchen Jin, and Huan Zhang. 2024. SVIP: Towards Verifiable Inference of Open-source Large Language Models. arXiv:2410.22307.
- [34] Florian Tramèr and Dan Boneh. 2019. Slalom: Fast, Verifiable and Private Execution of Neural Networks in Trusted Hardware. In *International Conference on Learning Representations (ICLR)*.
- [35] Chenkai Weng, Kang Yang, Xiang Xie, Jonathan Katz, and Xiao Wang. 2021. Mystique: Efficient Conversions for Zero-Knowledge Proofs with Applications to Machine Learning. In *30th USENIX Security Symposium*. USENIX Association, 501–518.
- [36] Tiancheng Xie, Tao Lu, Zhiyong Fang, Siqi Wang, Zhenfei Zhang, Yongzheng Jia, Dawn Song, and Jiaheng Zhang. 2025. zkPyTorch: A Hierarchical Optimized Compiler for Zero-Knowledge Machine Learning. Cryptology ePrint Archive, Paper 2025/535.
- [37] Jiaqi Xue, Yifei Zhao, Mengxin Zheng, Fan Yao, Yan Solihin, and Qian Lou. 2025. Securing Transformer-based AI Execution via Unified TEEs and Crypto-protected Accelerators. arXiv:2507.03278. Presents TwinShield.
- [38] Jianzhu Yao, Hongxu Su, Taobo Liao, Zerui Cheng, Huan Zhang, Xuechao Wang, and Pramod Viswanath. 2026. TAO: Tolerance-Aware Optimistic Verification for Floating-Point Neural Networks. In *Proceedings of the 21st European Conference on Computer Systems (EuroSys)*. ACM, 1515–1532. doi:10.1145/3767295.3803612
- [39] Dmytro Zakharov, Oleksandr Kurbatov, Artem Sdobnov, Lev Soukhanov, Yevhenii Sekhin, Vitalii Volovyk, Mykhailo Velykodnyi, Mark Cherepovskiy, Kyrylo Baibula, Lasha Antadze, Pavlo Kravchenko, Volodymyr Dubinin, and Yaroslav Panasenko. 2025. Bionetta: Efficient Client-Side Zero-Knowledge Machine Learning Proving. arXiv:2510.06784. arXiv version 2 (8 Dec 2025), whose Table 4 gives the numbers we use.
- [40] Hadas Zeilberger, Binyi Chen, and Ben Fisch. 2024. BaseFold: Efficient Field-Agnostic Polynomial Commitment Schemes from Foldable Codes. In *Advances in Cryptology – CRYPTO 2024 (LNCS, Vol. 14929)*. Springer, 138–169. doi:10.1007/978-3-031-68403-6_5
- [41] Zkonduit. 2026. EZKL: An Engine for Doing Inference for Deep Learning Models and Other Computational Graphs in a zk-SNARK. <https://github.com/zkonduit/ezkl>. GitHub repository, accessed September 2026.